COMPUTER AIDED ENGINEERING GRAPHICS

A9304

Experiment No. 4

Projection of Points - II

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Group Name

I B.Tech I Semester

Computer Science Engineering

VARDHAMAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

Affiliated to **JNTUH,** Approved by **AICTE**, Accredited by **NAAC** with **A++** Grade, **ISO 9001:2015** Certified

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Projection of Points II

1 Objectives

→ Understand the concept of orthogonal projections of points

→ Apply projection concepts for two points in different quadrants

→ employ the modify tools and dimension tools.

2 Pre-Lab Questions

Q1. If both the views of a point coincide with each other and lie below the reference line, state the angle in which the point lies.

Q2. State the similarities and dissimilarities in the projections of points which lie in the second angle and the fourth angle.

Q3. State the position of the point, the front view of which lies 50 mm below the reference line and the top view 30 mm above the front view.

Q4. State the position of the point, the top view of which lies 50 mm above the reference line and the front view 30 mm below the top view.

Q5. If the front view of a point lies above the reference line, state the possible angles in which the point may lie.

Q6. If the top view of a point lies above the reference line, state the possible angles in which the point may lie.

Q7. If the front view of a point lies below the reference line, state the possible angles in which the point may lie.

Q8. If the top view of a point lies below the reference line, state the possible angles in which the point may lie.

Q9. State the relationship between front view and top view of a point.

Q10. State the position of the point if its both views lie on the reference line.

3 Aim

Perform the following in AUTOCAD

- A point P is 15 mm above the H.P. and 20 mm in front of the V.P. Another point Q is 25 mm behind the V.P. and 40 mm below the H.P. Draw projections of P and Q keeping the distance between their projectors equal to 90 mm. Draw straight lines joining
  - their top views
  - their front views
- Two points A and B are in the H.P. The point A is 30 mm in front of the V.P., while B is behind the V.P. The distance between their projectors is 75 mm and the line joining their top views makes an angle of 45^0 with xy. Find the distance of the point B from the V.P.

4 Software Used

- AUTOCAD

5 System Requirements

1. **System Type:** Windows 10 64 Bit Operating System
2. **Processor:** i3
3. **RAM:** 4 GB

6 Commands Used

1. Units
2. Ucsicon
3. Limits
4. Line
5. Donut
6. Text
7. Arc
8. Trim
9. Erase
10. Move
11. Dimension Linear
12. Dimension Aligned
13. OSNAP
14. Ortho

7 Experimental procedure

1. Open the file with name *Template.dwg* from the local disk E and edit the title block.
2. Perform the units and ucsicon commands.
3. In the limits set the A4 sheet dimensions 297*X*210.

4. For the reference line XY use Line command and take some convenient distance of the line $(50mm)$.
5. Draw the projectors for point P and Q at a distance of 90 mm apart.
6. For point P (first quadrant): Mark front view (p') 15 mm above XY line and top view (p) 20 mm below XY line.
7. For point Q (third quadrant): Mark front view (q') 40 mm below XY line and top view (q) 25 mm above XY line.
8. Use donut command to mark all points (p', p, q', q).
9. Join the top views (p and q) with a straight line using the line command.
10. Join the front views (p' and q') with a straight line using the line command.
11. Use text command to label all views appropriately.
12. For the second problem, draw XY line and mark point A's top view (a) 30 mm below XY line.
13. Draw a line from (a) at 45° to XY line to locate point (b).
14. Project point (b) to find its distance from VP.
15. Use dimension commands to show all required distances.
16. Save the file in the local disk with appropriate name.

8 Outcomes

→ apply the dimension tools for the measure of unknown length

→ Application of various projection techniques within the CAD environment

→ Integration of projection techniques into the overall CAD modeling workflow, understanding how projections contribute to the design process.

9 Result

The solutions of the projection of points are analyzed and drawn in the AUTOCAD.

10 Post-Lab Questions

Q1. State the position of the point, the top view of which lies on the reference line and the front view 50 mm below it.

Q2. State the position of the point, the front view of which lies on the reference line and the top view 50 mm below it.

Q3. State the position of the point, the top view of which lies on the reference line and the front view 45 mm above it.

Q4. State the position of the point, the front view of which lies on the reference line and the top view 35 mm above it.

Q5. The line joining the front and top views of a point is called:

a) reference line
b) projector
c) connector
d) locus

Q6. The front view of a point is 40 mm above xy and the top view is 50 mm below xy. The point is:

a) 40 mm above H.P.
b) 40 mm below H.P.
c) 50 mm above H.P.
d) 50 mm below H.P.

Q7. State the position of a point the front view of which lies on the reference line and the top view is 40 mm above it:

a) 40 mm above H.P. and in the V.P.
b) 40 mm behind V.P. and in the H.P.
c) 40 mm below H.P. and in the V.P.
d) 40 mm in front of V.P. and in the H.P.

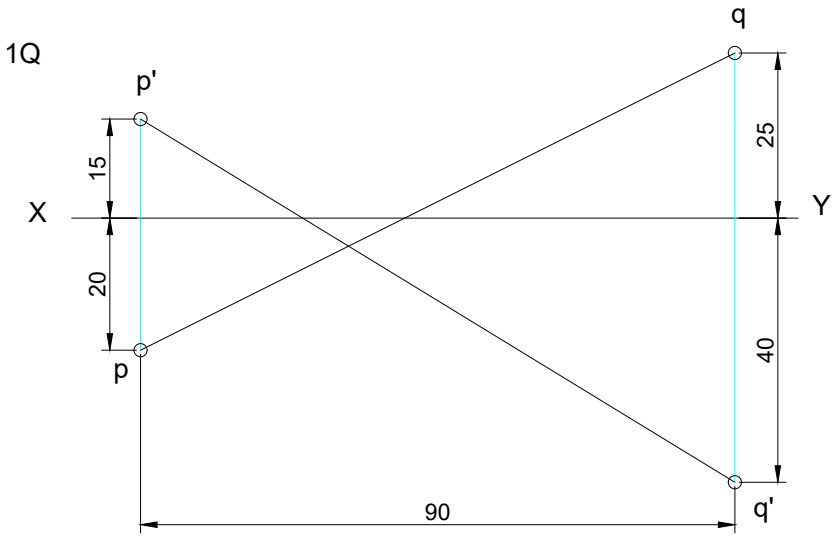
Q8. A point is 30 mm above H.P. and 40 mm behind V.P. Its front view will be:

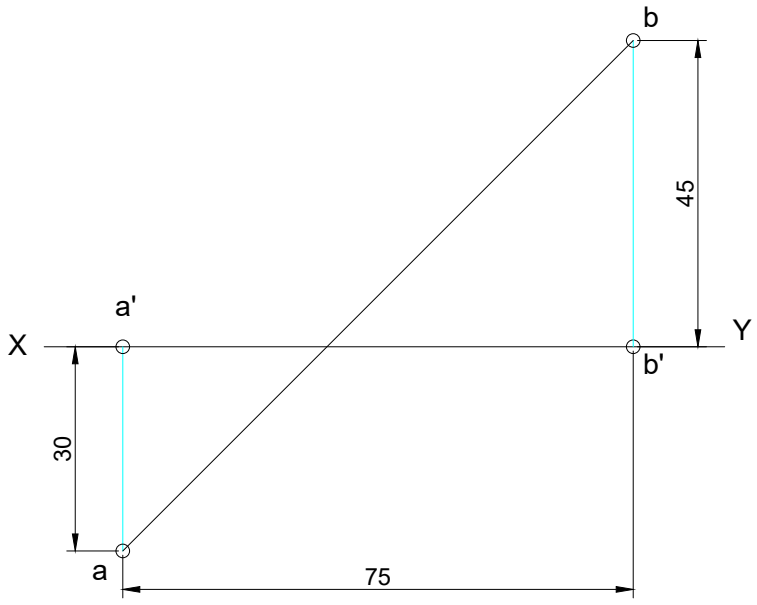
a) 30 mm above xy
b) 30 mm below xy
c) 40 mm above xy
d) 40 mm below xy

Q9. If a point lies in the fourth quadrant, its front view will be _____ the xy line and top view will be _____ the xy line:

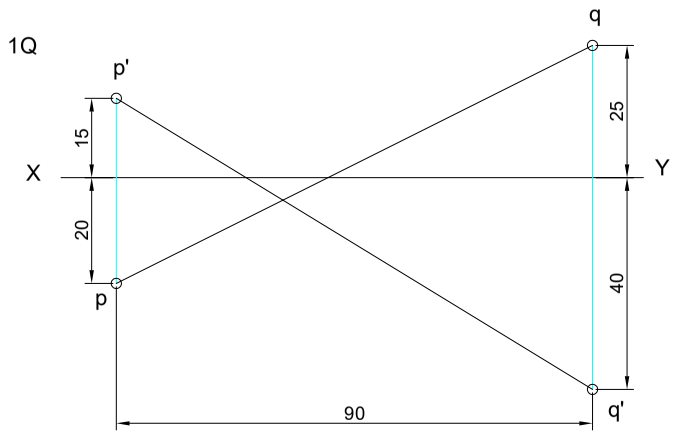
a) above, above
b) above, below
c) below, above
d) below, below

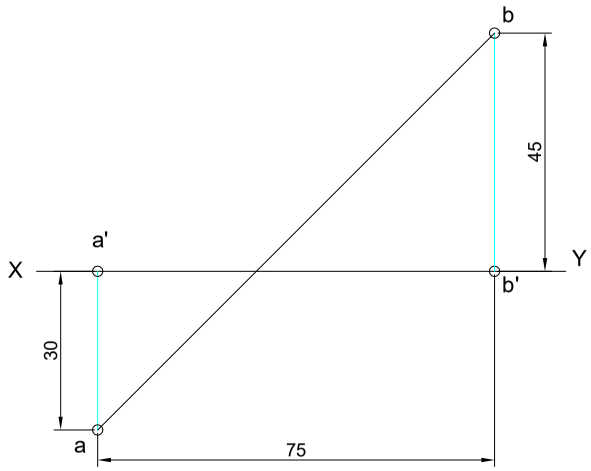
Solutions


1Q
q
p'
15
25
X
Y
20
40
p
90
q'


b
45
a'
X
Y
b'
30
a
75

OutPut





Perform the following in AUTOCAD

– A point P is 15 mm above the H.P. and 20 mm in front of the V.P. Another point Q is 25 mm behind the V.P. and 40 mm below the H.P. Draw projections of P and Q keeping the distance between their projectors equal to 90 mm. Draw straight lines joining

* their top views
* their front views

– Two points A and B are in the H.P. The point A is 30 mm in front of the V.P., while B is behind the V.P. The distance between their projectors is 75mmand the line joining their top views makes an angle of 450 with xy. Find the distance of the point B from the V.P.

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Exercise Problems

1. A point is 30 mm from the H.P. and 50 mm from the V.P. Draw its projections keeping it in all possible positions.

2. Draw the projections of the following points on a common reference line keeping the distance between their projectors 25 mm apart.

   (a) Point A is 40 mm above the H.P. and 25 mm in front of the V.P.

   (b) Point B is 40 mm above the H.P. and in the V.P.

   (c) Point C is 25 mm in front of the V.P. and in the H.P.

   (d) Point D is 25 mm above the H.P. and 30 mm behind the V.P.

   (e) Point E is in the H.P. and 30 mm behind the V.P.

   (f) Point F is 40 mm below the H.P. and 30 mm behind the V.P.

   (g) Point G is 25 mm below the H.P. and 40 mm in front of the V.P.

   (h) Point H is in the V.P. and 30 mm below the H.P.

3. Draw the projections of the following points on a common reference line keeping the distance between their projectors 30 mm apart.

   (a) Point P is 35 mm below the H.P. and in the V.P.

   (b) Point Q is 40 mm in front of the V.P. and 25mm below the H.P.

   (c) Point R is 45 mm above the H.P. and 20 mm behind the V.P.

   (d) Point S is 30 mm below the H.P. and 45 mm behind the V.P.

   (e) Point T is both in H.P. and V.P.

4. A point P is in the first quadrant. Its shortest distance from the intersection point of H.P., V.P and Auxiliary vertical plane, perpendicular to the H.P. and V.P. is 70 mm and its equidistance from principal planes (H.P. and V.P.) Draw the projections of the point and determine its distance from the H.P and V.P.

5. A point P is 20 mm below H.P. and lies in the third quadrant. Its shortest distance from xy is 50 mm. Draw its projections.

6. A point A is situated in the first quadrant. Its shortest distance from the intersection point of H.P., V.P. and auxiliary plane is 60 mm and it is eqidistant from the principal planes. Draw the projection of the point and determine its distance from the principal planes.

7. A point 30 mm above xy line is the plan-view of two points P and Q. The elevation of P is 45 mm above the H.P. while that of the point Q is 35 mm below the H.P. Draw the projections of the points and state their position with reference to the principal planes and the quadrant in which they lie.

8. A point Q is situated in first quadrant. It is 40 mm above H.P. and 30 mm in front ov V.P. Draw its projections and find its shortest distance from the intersection of H.P., V.P. and auxiliary plane